

Degree recovery from the ordinary loopy polynomial on a recognizable class of looped multigraphs

Xiaojian Zeng

PBC School of Finance, Tsinghua University, 100190

xiaojianzeng@foxmail.com

Abstract

We give an explicit formula recovering the degree multiset from the ordinary loopy polynomial for every finite multigraph with at most one loop at each vertex and at most one edge between any two loop-free vertices. Parallel edges incident to looped vertices are unrestricted. The formula combines three normalized derivatives with a quadratic correction that removes the two-vertex components counted by the first derivative in x_1 . It also recovers the joint distribution of loop count and non-loop degree. The hypotheses are recognizable from the polynomial, so an arbitrary multigraph with the same polynomial as a graph in this class has the same degree multiset. The unrestricted degree-recovery question of Kirillov–Nenashev–Shapiro–Vaintrob remains unresolved.

1 Scope and statement

All graphs are finite undirected multigraphs; disconnected graphs and isolated vertices are allowed. For a vertex v , let $\ell(v)$ be its number of loops and let $\delta(v)$ count its incident non-loop edges with multiplicity. Thus the degree is $d(v) = \delta(v) + 2\ell(v)$. For distinct vertices u, v , write $\mu(u, v)$ for the number of edges joining them.

We use the ordinary loopy polynomial $L_G(t, x_0, x_1, \dots)$ of [1, Definition 1.1]. It satisfies

$$L_G = L_{G/e} + tL_{G-e} \tag{1}$$

for every non-loop edge e , where G/e identifies its endpoints and *retains e as a loop*. It is multiplicative on disjoint unions, and the graph with one vertex and j loops has polynomial x_j ; also $L_\emptyset = 1$. Equalities of these invariants mean equality as full formal polynomials.

Kirillov, Nenashev, Shapiro and Vaintrob prove that L_G determines degrees for all loopless multigraphs [1, Theorem 4.3]. Their refined loopy polynomial recovers the joint distribution of $\ell(v)$ and $\delta(v)$ for arbitrary multigraphs [1, Theorem 6.15]. For the ordinary polynomial, a variable x_j can mark both a singleton with j loops and a larger forest component. Their Problem 11.4 asks whether L_G nevertheless determines the degree sequence when arbitrary loops are allowed. This note gives a partial result for the following class:

$$\mathcal{C} = \{G : \ell(v) \in \{0, 1\} \text{ for all } v, \quad \mu(u, v) \leq 1 \text{ whenever } \ell(u) = \ell(v) = 0\}. \tag{2}$$

In particular, $\mathcal{C}$ contains every simple non-loop core with optional single loops, and permits arbitrarily many parallel edges on any pair with a looped endpoint. There is no bound on the number of vertices or edges.

Theorem 1. Let $G \in \mathcal{C}$ have $n \geq 2$ vertices, and set $L = L_G$. Define

$$A(t) = (1-t)^{n-1} \frac{\partial L}{\partial x_0} \Big|_{x_i=(1-t)^{-1} \text{ for all } i}, \quad B(t) = (1-t)^{n-1} \frac{\partial L}{\partial x_1} \Big|_{x_i=(1-t)^{-1} \text{ for all } i}, \quad (3)$$

$$C(t) = \frac{(1-t)^{n-2}}{2} \frac{\partial^2 L}{\partial x_0^2} \Big|_{x_i=(1-t)^{-1} \text{ for all } i}, \quad Q(t) = \frac{A(t)^2 - A(t^2)}{2}. \quad (4)$$

All specializations are taken after differentiation; $A(t^2)$ is substitution into the resulting one-variable polynomial. Then

$$\boxed{\sum_{v \in V(G)} t^{d(v)} = A(t) + t^2 B(t) + t(Q(t) - C(t))}. \quad (5)$$

More precisely, the joint distribution is recovered by

$$\sum_{\ell(v)=0} t^{\delta(v)} = A(t), \quad \sum_{\ell(v)=1} t^{\delta(v)} = B(t) + \frac{Q(t) - C(t)}{t}, \quad (6)$$

where the division by t is exact in $\mathbb{Z}[t]$. Membership in $\mathcal{C}$ is determined by L_G . Consequently, if H is any finite undirected multigraph with $L_H = L_G$ and $G \in \mathcal{C}$, then $H \in \mathcal{C}$ and the two graphs have the same joint distribution and degree multiset. For the empty graph the degree polynomial is 0; for a one-vertex graph $L_G = x_{\ell(v)}$ gives it as $t^{2\ell(v)}$.

The substantive step is the cancellation in (5). The singleton-marking method is already present in [1, Proposition 4.4], and the recognition argument below is a special case of [1, Lemma 6.18]. The cancellation uses the full ordinary polynomial, including the nonlinear expression $A(t)^2 - A(t^2)$, to remove the nonsingleton contribution to B . This formula and the stated subclass theorem do not occur in the inspected version of [1]; no global priority claim is made. Reconstruction from the chromatic symmetric function of trees, as studied by Zeng [2], is adjacent graph-invariant reconstruction context. The result here concerns a different invariant and uses the argument below.

2 Marked forest components

Fix a linear order on the edges of G . A spanning forest here is any acyclic subset of non-loop edges, with all vertices retained; it need not be maximal. An edge outside a forest F is externally active if it is a loop, or if its endpoints lie in the same component and it is the least edge of its fundamental cycle. Let $\varepsilon_F(T)$ count the active edges assigned to the tree component T , and put $\varepsilon(F) = \sum_T \varepsilon_F(T)$. The forest expansion is

$$L_G = \sum_F t^{|E(G)| - |F| - \varepsilon(F)} \prod_{T \in \text{comp}(F)} x_{|V(T)| - 1 + \varepsilon_F(T)}. \quad (7)$$

This is [1, Theorem 2.2]. For completeness, choose the largest non-loop edge e . Forests omitting e have e inactive and contribute tL_{G-e} . Contracting e in a forest containing it gives a forest in G/e . The retained loop replaces the selected edge in the component index, while all other activities are preserved: deleting a largest edge from a fundamental cycle does not change its least edge. Thus these forests contribute $L_{G/e}$. For graphs with only loops the expansion is the product of the prescribed $x_{\ell(v)}$. This proves (7) by induction using (1).

Lemma 2 (Constant specialization). *For any multigraph K with N vertices, put $c = (1-t)^{-1}$ in $\mathbb{Q}(t)$. Then $L_K(t, c, c, \dots) = c^N$.*

Proof. This is [1, Lemma 4.2]. The loop-only case follows from multiplicativity. For a non-loop edge, deletion preserves N and loopy contraction reduces it by one; both reduce the number of non-loop edges. Induction gives $c^{N-1} + tc^N = c^N$. The argument includes $N = 0$. $\square$

Write $Z = \{v : \ell(v) = 0\}$ and $O = \{v : \ell(v) = 1\}$, and let $G - U$ denote deletion of the vertices in U and all incident edges. The index of a component in (7) is zero precisely when it is a singleton at a vertex of Z . Differentiation in x_0 marks one such component. Its incident non-loop edges are inactive, and deleting it preserves the fundamental cycles and activities in the remaining forest. Consequently

$$\frac{\partial L_G}{\partial x_0} = \sum_{v \in Z} t^{\delta(v)} L_{G-v}, \quad A(t) = \sum_{v \in Z} t^{\delta(v)}, \quad (8)$$

the second identity following from Lemma 2.

Marking two distinct zero-loop singletons removes $\delta(u) + \delta(v) - \mu(u, v)$ non-loop edges. The second derivative counts each unordered pair twice, so the same specialization gives

$$C(t) = \sum_{\{u, v\} \subseteq Z} t^{\delta(u) + \delta(v) - \mu(u, v)}, \quad (9)$$

$$Q(t) = \sum_{\{u, v\} \subseteq Z} t^{\delta(u) + \delta(v)}. \quad (10)$$

Here and below pair sums run over distinct vertices. In (10), $A(t)^2$ removes the diagonal from $A(t)^2$, and division by two removes the ordering.

Let E_Z be the set of unordered adjacent pairs in Z , and put

$$S(t) = \sum_{\{u, v\} \in E_Z} t^{\delta(u) + \delta(v) - 2}. \quad (11)$$

Its exponents are nonnegative. By the hypothesis on Z , each pair in E_Z has multiplicity one. Nonadjacent pairs contribute equally to (9) and (10); hence

$$C(t) - Q(t) = t(1 - t)S(t). \quad (12)$$

There are precisely two ways for a component in (7) to contribute x_1 . It can be a singleton at a vertex in O , or a two-vertex tree with no externally active edges. Under (2), the second possibility consists of the unique edge of a pair in E_Z . Indeed a larger tree has index at least two, and a loop at either endpoint of a two-vertex tree raises its index to at least two.

Deleting a marked singleton in O contributes $t^{\delta(v)} L_{G-v}$: its loop is removed from both the total edge count and the activity count, and thus adds no power of t . For a marked two-vertex tree uv , deleting its vertices removes $\delta(u) + \delta(v) - 1$ edges and one selected forest edge. The resulting factor is $t^{\delta(u) + \delta(v) - 2} L_{G - \{u, v\}}$. The remaining forest has $n - 2$ vertices, so Lemma 2 yields

$$B(t) = H_1(t) + (1 - t)S(t), \quad H_1(t) = \sum_{v \in O} t^{\delta(v)}. \quad (13)$$

Combining (12) and (13) proves $H_1 = B + (Q - C)/t$. Finally the degree polynomial is $A + t^2 H_1$, which proves (5) and (6).

These formulas are polynomial identities, including at $t = 0$ and $t = 1$. To see this directly, a monomial of ordinary x -degree $k \leq n$ contributes an integer multiple of $t^p(1 - t)^{n-k}$ to each normalized derivative A, B, C . For C , the integer multiplier is the binomial coefficient choosing two x_0 factors. Equation (10) gives $Q \in \mathbb{Z}[t]$, and (12) justifies the exact division by t . Thus the use of $\mathbb{Q}(t)$ introduces no exceptional specialization.

3 Recognizing the class

Proposition 3. *The ordinary loopy polynomial determines membership in $\mathcal{C}$. If its loop multiset consists of a zeros and b ones, and it has M non-loop edges, then for $a \geq 2$ and $M \geq 2$ the coefficient*

$$[t^{M-2}x_0^{a-2}x_1^bx_2]L_G \quad (14)$$

counts the unordered pairs of zero-loop vertices with at least two parallel edges.

Proof. The unique term of maximum ordinary x -degree in (7) comes from the empty forest and is

$$t^M \prod_{v \in V(G)} x_{\ell(v)}. \quad (15)$$

It determines the vertex count, M , and the multiset of loop counts, as in [1, Proposition 2.9]. Thus the first condition of (2) can be tested directly.

Assume this condition holds. A term of ordinary x -degree $n - 1$ comes from a forest containing a single non-loop edge uv . If this edge is the i th smallest in its parallel class, exactly $i - 1$ other edges in that class are active. Its two-vertex component contributes $x_{\ell(u)+\ell(v)+i}$ and the power of t is $M - i$; the other components contribute $\prod_{w \neq u,v} x_{\ell(w)}$. This is the relevant instance of [1, Lemma 6.18].

The power in (14) forces $i = 2$. If either endpoint has a loop, the component index is at least three, so the target monomial cannot occur. If both endpoints have no loops, exactly the second-smallest edge contributes that monomial, provided the pair has multiplicity at least two. This proves the asserted count. All coefficients in the forest expansion are nonnegative, so its vanishing is equivalent to the second condition in (2). If $a < 2$ or $M < 2$, that condition is automatic. The empty and one-vertex cases follow directly from their polynomials. $\square$

If $G \in \mathcal{C}$ and $L_H = L_G$, Proposition 3 first puts H in $\mathcal{C}$. Applying the same derivatives to their common polynomial then gives the same joint distribution and degree multiset. This completes Theorem 1.

4 Boundary and exact checks

The restriction on parallel edges cannot simply be dropped from the displayed formula. Two zero-loop vertices joined by two parallel edges have

$$L = t^2x_0^2 + tx_1 + x_2.$$

For this graph, $A = 2t^2$, $B = t(1 - t)$, $C = t^2$, and $Q = t^4$. The right side of (5) is $2t^2 - t^4 + t^5$, whereas the degree polynomial is $2t^2$. This is a failure of the formula outside its hypotheses. It does not give two graphs with equal loopy polynomials and different degree multisets. In fact the source's loopless theorem already recovers the degrees of this example.

The general proof above is supplemented by exact finite checks. A first implementation checked the identity on 68,747 graph instances: every simple non-loop core with 0/1 loops through five vertices, and every four-vertex 0/1-loop assignment with multiplicities at most one on zero-loop pairs and at most three on all other pairs. These totals count test instances and include overlap between the two families. An additional 168 generated graphs on two to eight vertices were checked using deletion–contraction and forest expansions in opposite edge orders; 87 belonged to $\mathcal{C}$. These generated graphs are not an exhaustive box.

A separately written verifier, importing no code from the first implementation, checked class recognition on 2,152 graph instances. These comprised the 168 additional inputs; all 1,728 three-vertex graphs with loop counts 0, 1, 2 and multiplicities 0, 1, 2, 3; all 176 two-vertex graphs with loop counts 0, 1, 2, 3 and multiplicities $0, \dots, 10$; five empty or one-vertex inputs; 72 generated members of $\mathcal{C}$ on four to nine vertices; and three high-multiplicity inputs, including 40 parallel edges. There were 560 instances in $\mathcal{C}$. For those with at least two vertices, the degree identity passed 7,256 exact integer evaluations, including $t = 0$ and $t = 1$. For each graph the difference has degree at most $\max(2M + 1, M + 2)$, and the verifier used more than that many distinct points, thereby certifying the polynomial identity for that input. No finite test is used to infer the unbounded theorem.

Reproducibility and assistance. The accompanying computational dossier contains the initial verifier, the forest-expansion comparison, the separately implemented deletion–contraction verifier, and their exact input and output records. The separate verifier eliminates a parallel class of size r by either deleting all its edges with weight t^r , or contracting when i edges remain with weight t^{r-i} and retaining i loops, for $1 \leq i \leq r$. AI assistance was used to develop and check the argument, implement the computations, and draft this note. A separate AI review reconstructed the proof and wrote the second verifier. This is a separate checking pass, not external peer review or formal proof-assistant certification.

Remaining scope. The theorem recovers degrees and the joint loop/non-loop degree distribution on exactly (2); it does not reconstruct graph isomorphism types. No conclusion here settles arbitrary loop multiplicities or the mixed looped case with parallel edges between zero-loop vertices. The unrestricted question in [1, Problem 11.4] remains unresolved by this work. The source comparison is with the inspected first arXiv version; broader historical equivalence and priority screening remains incomplete.

References

- [1] A. Kirillov, G. Nenashev, B. Shapiro, and A. Vaintrob, *The Loopy Polynomial: from Tutte’s Universal V -Function to Bizonotopal Geometry*, arXiv:2609.07728v1, 2026. <https://arxiv.org/abs/2609.07728v1>.
- [2] Zijian Zeng, *Multiplicity-Isolated Cores and Chromatic Symmetric Reconstruction of Trees*, arXiv:2608.18179, 2026. <https://doi.org/10.48550/arXiv.2608.18179>.